scientific reports

Check for updates

OPEN

Progressive multi-scale multi-attention fusion for hyperspectral image classification

Hu Wang[1,2], Sixiang Quan[3], Jun Liu[1,2]✉, Hai Xiao[3], Yingying Peng[1,2], Zhihui Wang[1,2] & Huali Li[4]

In recent years, due to the unique spatial-spectral characteristics of hyperspectral images, they have played a crucial role in many fields. The effective extraction of features using deep neural networks, followed by the design of efficient and high-precision network algorithm structures, has gradually become a research hotspot. Hyperspectral images are difficult to obtain and have limited samples. Although hyperspectral image classification methods based on convolutional neural networks (CNN) have noticeably improved performance, there are still certain shortcomings in the extraction of detailed and local features. Therefore, how to fully utilize spatial and spectral information in situations with limited samples has become a challenging problem. To address this issue, inspired by the PID controller, this paper proposes a Progressive Multi-Scale Multi-Attention Fusion (PMMF) network structure that simultaneously extracts features from the Proportional (P), Integral (I), and Derivative (D) branches. The complementary responsibilities of the three branches address the issue of feature loss in details and improve the network's learning efficiency across feature maps of different scales. By cleverly extracting features from different branches multiple times, the fusion of multi-scale features is achieved, avoiding the limitations of single-scale feature representation. The proposed multi-attention fusion module applies the most suitable attention mechanism according to the representation form of each branch, fully extracting features from each branch, enriching the information contained in the feature maps, and greatly enhancing the classification accuracy of hyperspectral images.

Over the past few decades, the rapid development of hyperspectral technology has led to its widespread application in many fields, such as vegetation[1], soil properties[2], environmental monitoring[3], geophysical exploration[4,5], and the military[6]. Hyperspectral images, rich in spectral and spatial information, distinguish physical differences between surface materials better than natural images through extensive and dense spectral imaging[7]. This makes hyperspectral imaging a highly dynamic research field. Hyperspectral technology has also received extensive attention in the field of remote sensing[8]. In remote sensing, hyperspectral data can provide more detailed spectral information for the description of ground objects, and this continuous spectral information can be used to image ground or urban surfaces[9]. Therefore, hyperspectral data is widely used in land cover classification tasks and has already achieved good classification results. Due to the high resolution and integrated spectrum-image nature of hyperspectral images, many researchers have conducted extensive studies on their classification algorithms. Hyperspectral image classification methods can mainly be divided into two categories: traditional methods and deep learning-based methods.

Hyperspectral image classification aims to assign a unique category label to each pixel in the image. Traditional hyperspectral image classification typically involves feature extraction followed by the use of a classifier to categorize the hyperspectral image. In the early stages of hyperspectral image classification research, many spectral-based feature extraction methods were proposed, such as Support Vector Machine (SVM)[10], Multinomial Logistic Regression (MLR)[11], Random Forest[12], Random Subspace[13], Sparse Representation[14], and K-Nearest Neighbors (KNN)[15]. These traditional classification methods, which only utilize spectral features, are computationally simple and easy to implement. Among them, the KNN algorithm is satisfactory due to its theoretical simplicity; SVM exhibits superior classification performance in small sample problems and large dataset classification tasks and is widely used in hyperspectral image classification tasks. Moreover, some feature extraction or dimensionality reduction methods have also attracted attention, such as Independent Component

[1]School of Informatics, Hunan University of Chinese Medicine, Changsha 410208, China. [2]AI TCM Lab Hunan, Changsha 410208, China. [3]Second Surveying and Mapping Institute of Hunan Province, Changsha 430103, China. [4]College of Electrical and Information Engineering, Hunan University, Changsha 410082, China. ✉email: jun.liu@hnucm.edu.cn

Analysis (ICA)[16], Principal Component Analysis (PCA)[17], and Linear Discriminant Analysis (LDA)[18]. However, the classification results of the aforementioned pixel-based classifiers are not satisfactory. To better classify hyperspectral images, some effective spatial-spectral feature representation methods have been proposed[19,20]. With the large-scale increase in data and the complexity of application scenarios, traditional methods reveal their limitations, as they mainly rely on spectral information for feature learning, focusing on extracting shallow features and neglecting the spatial correlation of pixels, resulting in poor discrimination of ground object features. Pixels in spatially adjacent regions are very likely to belong to the same category, and fully utilizing the spatial features of hyperspectral images can further improve classification results. To optimize the extraction of spatial features from images, various morphological profile-based operations have been developed, such as Morphological Profiles (MPs)[21], Extended MPs (EMPs)[22], and Extended Multi-Attribute Profiles (EMAP)[23]. The features extracted by mathematical morphology operators like EMPs heavily rely on prior information, leading to poor adaptability to different datasets and insufficient robustness in complex environments. Additionally, the discriminative power of the extracted shallow features is inadequate. Relying solely on either spectral information or spatial information will inevitably result in suboptimal classification outcomes.

In recent years, deep learning has rapidly developed and been applied in many fields such as image processing[24] and natural language processing[25]. Using deep learning for hyperspectral image classification has become a hot research topic. In contrast to traditional machine learning-based classification approaches, deep learning methods possess significant advantages in feature extraction. They are capable of delving into the intrinsic abstract features of data and more effectively uncovering the complex internal structure of high-dimensional data. Thereby improving image classification accuracy. Lin et al. proposed using autoencoders for hyperspectral image classification[26], presenting a deep learning method that integrates spectral and spatial features. Zheng et al. used Deep Belief Networks (DBN) to classify EEG emotions[27]. Both SAE and DBN require transforming data into one-dimensional vectors to accommodate their network structures, resulting in the inability to consider the spatial information of images during feature extraction. Convolutional Neural Networks (CNN) can directly process two-dimensional or three-dimensional images, effectively alleviating the aforementioned issues. Li et al.[28] used 1-D-CNN for ECG signal classification. Qi Wang et al. designed a 2-D-CNN[29], effectively learning discriminative features from multi-source data and improving the generalization capability of CD algorithms. To address the limitations of 2-D-CNN, Li et al.[30] proposed a 3-D-CNN to jointly extract spatial and spectral features. To better capture spatial-spectral information, Roy et al. proposed a hierarchical network that combines features extracted by 3D-CNN and 2D-CNN[31]. Manna et al. proposed an improved residual network to capture spectral and spatial features using an end-to-end training method[32]. CNN-based models have the unique advantage of feature representation among different channels and are proficient at extracting local features. However, their receptive fields are influenced by the size of their convolution kernels, limiting their ability to extract and represent complex spatial and global features. Lei et al. proposed an expanded CNN model[33], constructed by replacing the traditional CNN's convolution kernels with dilated convolution kernels, and tested the expanded CNN model on the MNIST handwritten digit recognition dataset. Huang proposed a DenseNet-based CNN[34], which enhanced feature propagation and achieved better classification performance by introducing dense connections into the network. Zhang et al. proposed an efficient remote sensing image scene classification structure[35], named cnn-capsnet, to fully exploit the advantages of both CNN and Capsule Network (CapsNet) models. Since the features obtained by single-scale convolution kernels are not sufficiently rich, many methods based on multi-scale convolution kernels have been used to extract more abundant features, thereby improving the classification performance of HSIs[36–38].

Attention mechanisms have also shown great potential in the field of computer vision. In cognitive science, humans tend to focus on more important information and ignore other information. Attention mechanisms can be seen as a mimicry of human vision and have been widely applied in many areas of computer vision[39–41]. Fu et al. proposed a Dual Attention Network (DANet)[42] to extract spatial-spectral information. Ma et al. proposed a Dual-Branch Multi-Attention Network (DBMA)[43] and achieved promising classification results. To further improve the classification performance of HSIs, Li et al. proposed a Dual-Branch Dual Attention Network (DBDA)[44]. Cui et al. proposed a novel Dual-Triple Attention Network (DTAN)[45], which can effectively classify hyperspectral images by capturing cross-dimensional interactive information. Tang et al. proposed a model incorporating Graph Attention[46], which allows information flow to achieve reasonable bit allocation. Liu et al. proposed a Spatial-Channel Attention mechanism[47], which captures cross-dimensional interactions without involving dimensionality reduction and brings significant performance improvements with negligible computational overhead.

Hyperspectral image classification with deep learning has become a popular research topic; existing methods have achieved good classification performance but still exhibit certain limitations. For example, (1) single-scale features are insufficient: traditional 1-D/2-D/3-D CNNs rely solely on fixed-size convolution kernels, making it difficult to simultaneously capture fine-grained textures and global semantics and leading to overfitting. (2) Cross-scale information transfer is weak: although current attention networks introduce multiple attention mechanisms, feature fusion mainly occurs at a single scale, resulting in the loss of discriminative information. (3) Computational and feature redundancies coexist: DenseNet and multi-scale convolutions improve accuracy but bring parameter inflation and redundant features, which are unfavorable for edge-device deployment. In order to enhance image classification accuracy, how can one effectively capture features at various scales within an image and select those that contribute to improved classification performance? And how can the flow of information and efficient transmission between different feature maps within a network model be strengthened? This paper integrates these concerns into its network model design, with the aim of boosting the precision of image classification through these methods:

(1) To alleviate the loss and insufficiency of single-scale features, inspired by the complementary three-branch responsibilities of a PID controller, this paper propose a Progressive Multi-scale Multi-attention Fusion (PMMF) network for HIC. The network adopts an "extract-stack-extract-fuse" paradigm, which repeatedly extracts features from different branches and then merges them. This design elegantly fuses multi-scale features, avoids the limited expressiveness of single-scale representations, and enriches the resulting feature maps with more information.
(2) To capture features more effectively and emphasize important ones, this paper introduce a Shortcut Weight Channel Attention (SWCA:Shortcut Weight Channel Attention) module. By connecting features across scales via shortcut pathways, SWCA refines the representations and simultaneously captures fine-grained textures and global semantics, enabling the network to focus on salient features and improving overall performance.
(3) When fusing features from the P, I, and D branches, this paper address weak cross-scale information transfer and the risk of information loss or redundancy by proposing a Multi-attention Fusion Module. Each branch is assigned a distinct attention mechanism tailored to its specific characteristics, ensuring that features within every branch are fully exploited and leading to more effective image classification.

The remainder of this paper is structured as follows. The basic structure and principles of the PMMFNet, and details the progressive multi-scale multi-attention network architecture are described in Section "Proposed method", while the comparative experiments are presented in section "Experimental setup and results". We summarize the review of image classification, as well as prospects for future work in the last Section "Conclusion".

Proposed method

Spatial shuffle sample preprocessing

CNNs have a large number of parameters and require a significant amount of samples for training, while the small sample characteristics of hyperspectral images easily lead to overfitting. Among the many small-sample learning algorithms, spatial-spectral fusion is a commonly used technique. For a given sample pixel, the pixels within an $N \times N$ neighborhood around that pixel are selected and combined as a single sample. The spatial and spectral features of this sample are then extracted and fused before being fed into a designed classification algorithm. Based on the spatial structure of neighboring pixels, a spatial shuffle scheme for small-size patches has been proposed[48]. Using this approach with a basic CNN architecture can achieve relatively high classification accuracy.

Specifically, for an $N \times N$ neighborhood size, excluding the central pixel, there are $N \times N - 1$ other pixels. Keeping the position of the central pixel unchanged, the remaining $N \times N - 1$ pixels are shuffled randomly to obtain a new sequence, as illustrated in Fig. 1.

In this study, specifically when $N = 5$, there are a total of $24! = 6.2e{+}23$ potential patterns. With M samples, this results in $M \times (N \times N - 1)!$ samples, significantly increasing the sample quantity. Despite the continued presence of substantial similarities between samples, these patterns represent potential distributions of real-world features. They also provide deep learning models with greater learning capacity.

Overall framework

The network proposed in this paper is similar to a PID controller. A PID controller consists of three components: Proportional, Integral, and Derivative, as shown in Fig. 2-Upper. PID is a type of analog closed-loop control method with no steady-state error. It is an effective control method that not only eliminates the steady-state error present in deviation control but also addresses stability and responsiveness. The goal of the PID control algorithm is to adjust the controller output based on the control error (i.e., the difference between the desired value and the actual value) to bring it as close as possible to the desired value. The proportional term reflects the current magnitude of the error, the integral term considers the accumulation of the error, and the derivative term indicates the rate of change of the error. By adjusting the values of these three parameters, the stability and response speed of the system can be controlled.

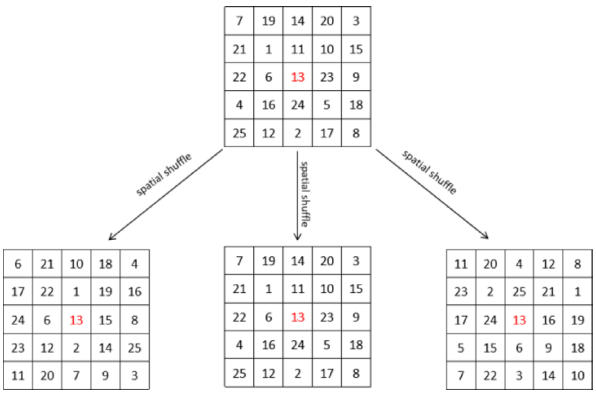


Fig. 1. Schematic diagram of spatial shuffle, where red font indicates the center pixel.

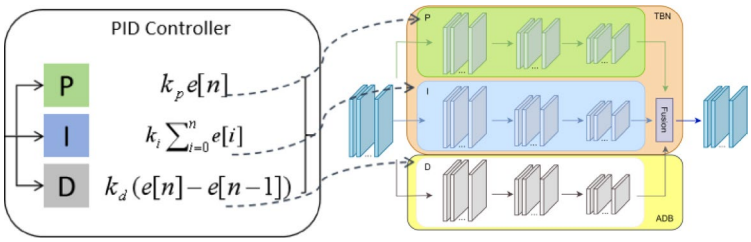


Fig. 2. The analogy between the PID controller and the proposed network.

The P controller focuses on the current signal, while the I controller accumulates all past signals. Due to the cumulative inertia effect, when the signal reverses, the output of a simple PI controller tends to overshoot. Therefore, a D controller is introduced; if the signal decreases, the D component becomes negative, acting as a damper to reduce the overshoot. The implementation of a PI controller can be written as:

$$C_{\text{out}}[n] = k_p e[n] + k_i \sum_{i=0}^{n} e[i] \tag{1}$$

The context and detailed information are parsed separately through multiple convolutional layers, with both the detailed branch and the context branch consisting of 3 layers, without BN (Batch Normalization) and ReLU. The D controller reduces overshoot by making the control output sensitive to changes in the input signal. The detailed branch parses all types of semantic information, even if not very accurately, while the context branch aggregates low-frequency contextual information and is semantically similar to the operation of using a large average filter. When the system is disturbed (including setpoint changes and disturbances) and the controlled variable deviates from the control value, the PID controller can make the system automatically return to the set value in a stable and fast manner. Simply put, PID stands for Proportional-Integral-Derivative control. When a system is disturbed (including changes in the setpoint and disturbances), causing the controlled variable to deviate from the set value, the PID controller can automatically and quickly stabilize the system and bring it back to the set value.

The proposed PMMFNet in this paper is inspired by the excellent three-branch structure of the PID controller. The relationship between the P/I/D controller and the three branches is as follows: The I, acting as the integrator of all signals in the PID, corresponds to the main trunk branch of this network, which integrates local and global context information to parse remote dependencies. The P, focusing on the current signal in the PID, corresponds to the proportional branch (P), which analyzes and preserves detailed information from the low-level high-resolution feature maps in advance. The D, serving as a damper in the PID, corresponds to the derivative branch (D), which extracts deeper high-frequency features to predict boundary areas. On this basis, the SWCA (Shortcut Weight Channel Attention) module is designed and added to each branch of the network. This module reduces internal covariate shift, allows for a larger learning rate to accelerate convergence, activates nonlinear capabilities, and improves computational efficiency. Finally, the P, I, and D branches are fused together, merging the branches with different responsibilities to produce the final output, noticeably improving accuracy. The overall structure is shown in the Fig. 2.

As shown in Fig. 3, the network structure proposed in this paper integrates the SWCA module and branch fusion to enhance the feature extraction capability from images. Assuming the hyperspectral image has D bands and spatial dimensions W × H, the overall computational load of the model is reduced after spatial shuffle sample preprocessing. This preprocessing step removes redundant information and noise while preserving essential spectral details. Since each input band is different, all bands less than 200 are filled with zeros. The final input size of feature maps fed into the network is consistently maintained at 25 × 200.

In the overall network architecture, a SWCA module is introduced during the feature extraction stage of each branch. The I branch serves as the main backbone, while inputs progressively differ for the other two branches. All three branches employ progressive multi-scale inputs and convolutional feature extraction with varying kernel sizes. This approach enhances feature propagation, noticeably improving feature representation by effectively capturing less prominent local features and thereby boosting classification accuracy. Furthermore, different branches aggregate features from different scales and utilize distinct attention mechanisms to further extract and learn features, facilitating effective feature reuse. Finally, the rich features obtained from the three branches are combined and passed through fully connected layers to accomplish the classification task.

The SWCA module

The SWCA proposed in this paper is described in the following figure.

The SWCA module is shown in Fig. 4, the feature map is first convolved, followed by batch normalization and activation, resulting in a new feature map. A skip connection is then used to concatenate the original feature map with the new one, enhancing the feature representation. Subsequently, a max pooling operation is applied to generate a new input $F_{H\times W\times C}$. The obtained input $F_{H\times W\times C}$ is first processed with a global average pooling and then stretched to obtain input $x = [\times 1, \times 2, \ldots, xm]$(where m represents the batch size). This approach can reduce internal covariate shift, enabling the use of larger learning rates to speed up convergence. The activation introduces

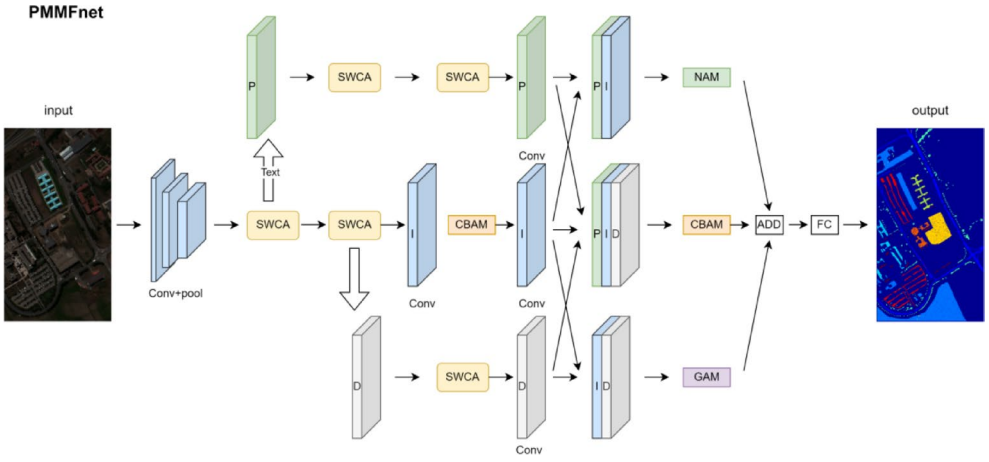


Fig. 3. The proposed progressive multi-scale multi-attention fusion network architecture of the Proportional-Integral-Derivative (PID) network.

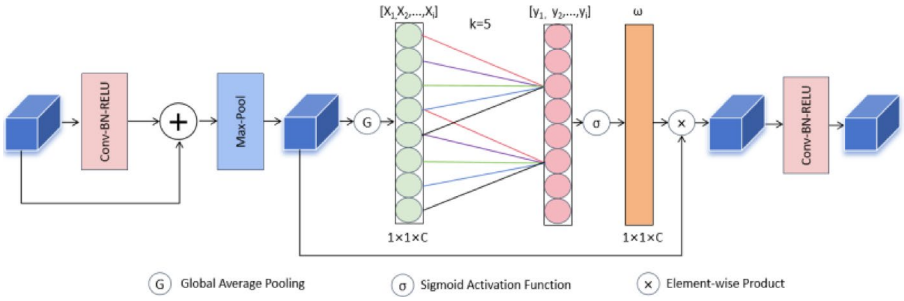


Fig. 4. Operational flow of the SWCA module.

non-linearity, which aids in learning complex features. Moreover, the subsequent pooling operation reduces dimensionality and extracts key features, improving computational efficiency and robustness. The calculation of the mean A_B and variance D^2_B of x is performed, followed by normalization to obtain x_i. A 1-dimensional convolution operation with a kernel size of k is then carried out, during which learnable parameters α and β are introduced. The comprehensive formula yi, incorporating the aforementioned steps, is as follows:

$$D_B^2 = \frac{1}{m}\sum_{i=1}^{m}(x_i - A_B)^2 \tag{2}$$

$$x_i = \frac{x_i - A_B}{\sqrt{D_B^2 + \varphi}} \tag{3}$$

$$y_i = \alpha\frac{x_i - A_B}{\sqrt{D_B^2 + \varphi}} + \beta \tag{4}$$

$$\omega = \frac{1}{1+e^{-x}}C1D_k(y) \tag{5}$$

The C1D denotes a one-dimensional convolution that involves k parameters, and after passing through the σ activation function, it obtains the weight ω. After obtaining the weights, they are multiplied element-wise with the original feature maps to produce the output feature maps. Each element of the new feature maps, obtained after one convolution, normalization, and activation, is added to the corresponding element of the original feature maps, fully representing the features to obtain the final feature maps.

Multi-attention branch fusion

The multi-attention branch fusion structure proposed in this paper is shown in Fig. 5.

Through the integration of three different attention mechanisms, the fusion process involves both feature map addition and concatenation. Different branch configurations employ varying sizes of convolutional kernels and pooling layers to extract features at different scales, effectively capturing local details and global contextual information. The network's branch structure enables parallel learning of diverse feature representations,

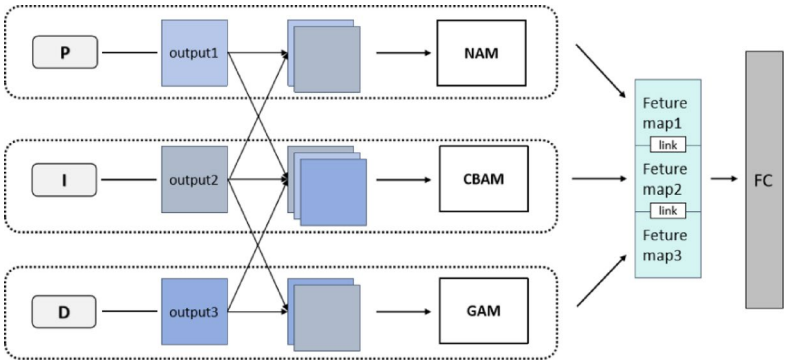


Fig. 5. Operational flow of the multi-attention branch fusion.

enhancing efficiency and performance. The I branch remains the main branch, with the features from the P and D branches integrated into it, while the PD branches serve as secondary branches with the features from the I branch also integrated into them. This approach increases the non-linearity of the network, helps the network learn more complex features, and allows it to capture the feature representations from different branches. This study employs three different attention mechanisms:Normalization-based attention module (NAM)[49], Convolutional block attention module (CBAM)[50], and Global attention mechanism (GAM)[51]. The NAM attention mechanism primarily functions to automatically learn the significance weights of various channels, suppress noisy channels, enhance effective feature channels, highlight key spatial regions in the feature map, and attenuate background interference. It is suitable for the functionality of branch P, which is designed to pre-save detailed information in the high-resolution feature maps of shallow layers and then conduct feature extraction targeting the effective information. The CBAM attention mechanism mainly works through the collaborative effort of channel and spatial attention, which noticeably enhances the representation capability of convolutional neural networks. It is appropriate for branch I, which aggregates local and global context information to parse long-range dependencies. The GAM attention mechanism mainly serves to improve the neural network's understanding of complex visual patterns. It corresponds to branch D's extraction of deeper-level high-frequency features for boundary region prediction. After conducting multiple sets of different attention-matching tests, due to the different scales of feature extraction and the manifestations of features in the three branches, the attention mechanism most suitable for each branch was selected respectively to further strengthen the feature representation. The experimental results also proved the correctness of the above theoretical analysis. Finally, the feature maps obtained from the three branches were flattened and concatenated, and then transformed into the final classification prediction results through the fully connected layer.

NAM attention branch (P)

The Derivative (D) branch extracts high-frequency features to predict boundary regions.

Figure 6 illustrates the detailed structure of the NAM (P) branch, which includes two SWCA modules and a series of convolutional layers. Finally, the output is obtained by processing the feature map through the NAM attention mechanism. The input is the output obtained after the initial SWCA processing of the original feature map, which is still at the shallow stage of feature extraction. Within the NAM (Neighborhood Attention Module) branch, two instances of SWCA modules are utilized for feature extraction, followed by two 1×3 convolutions (CNN) to regularize the feature map size (M ×M). Subsequently, it undergoes six times of CNN with spatial sizes of 7×7, 7×7, 4×4, 3×3, 3×3, and 3×3 for multi-scale spatial feature extraction. Following this, it goes through a channel-spatial attention mechanism to obtain the output, which is then stacked with the output from the CBAM (Convolutional Block Attention Module) branch. Ultimately, the output undergoes the most suitable NAM attention mechanism to obtain the final output.

The NAM, as shown in the Fig. 6a and b of the diagram above, is an efficient and lightweight attention mechanism that integrates modules from CBAM and redesigns channel and spatial attention sub-modules. In the channel attention sub-module, The scaling factor from batch normalization is utilized, as shown in the formula below. This scaling factor measures the variance across channels and indicates their importance.

$$B_{out} = BN(B_{in}) = \gamma \frac{B_{in} - \mu_B}{\sqrt{\sigma_B^2 + \varepsilon}} + \beta \tag{6}$$

where μ_B and σ_B are the batch standard deviations of B, γ and β are trainable affine transformation parameters (scale and shift). The NAM channel attention sub-module (Channel Attention Module) is shown in Fig. 6a above, where the output feature is $M_S = \text{sigmoid}(W_\gamma(BN(F_1)))$, with $W_\gamma = \frac{\gamma_i}{\sum_{j=0} \gamma_j}$ representing the weights.

Applying the scale factor of Batch Normalization to the spatial dimension is called Pixel Normalization. Correspondingly, in the NAM spatial attention sub-module (Spatial Attention Module) as shown in Fig. 6b

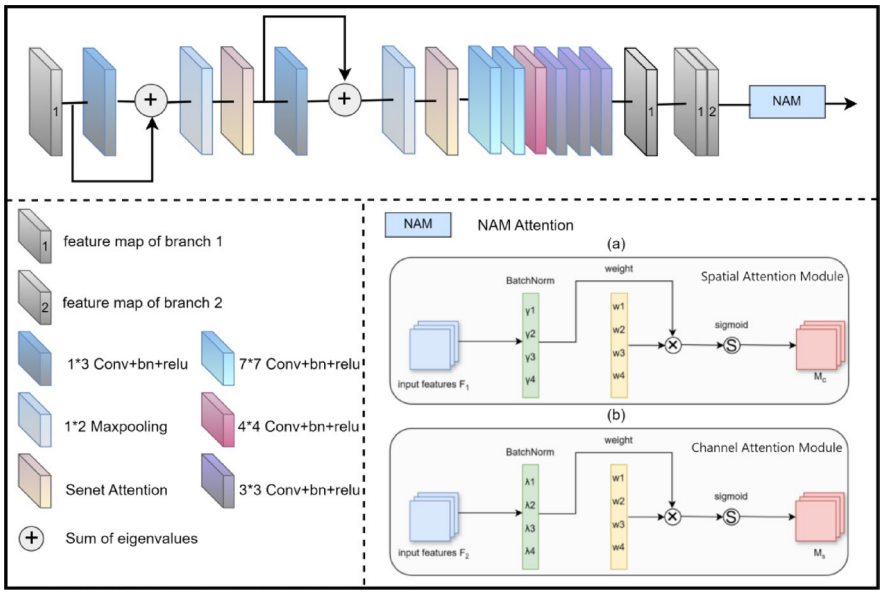


Fig. 6. Network structure of the NAM (P) branch.

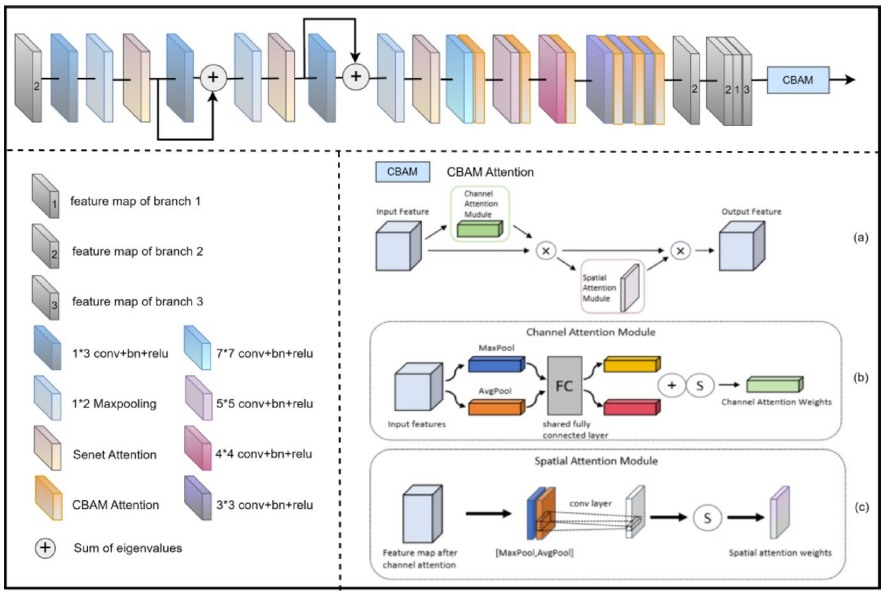


Fig. 7. Network structure of the CBAM (I) branch.

above, the output feature is $M_C = \text{sigmoid}(W_\gamma(BN(F_2)))$, where $W_\gamma = \frac{\lambda_i}{\sum_{j=0} \lambda_j}$ represents the weights, The scaling factors γ and λ are associated with the two sub-modules of channel and spatial attention, respectively.

CBAM attention branch (I)

The Integral (I) branch aggregates contextual information locally and globally to resolve long-range dependencies.

Figure 7 shows the detailed structure of the CBAM (I) branch, which is also the main backbone branch of the network throughout the entire feature extraction process, which involves two consecutive SWCA modules for feature extraction. The branch makes use of three 1×3 CNNs to standardize the feature maps to an M by M dimension. Additionally, it employs six convolutional layers with dimensions 7×7, 5×5, 4×4, 3×3, 3×3, and 3×3 each for the coalesced extraction of spatial features across multiple scales. However, to enhance feature propagation and significantly improve feature representation, after each CNN feature extraction following the reshaping step, the CBAM branch integrates a CBAM module. This addition slightly increases computational and parameter overhead but greatly enhances model performance. As the primary branch of the network, the CBAM branch combines the outputs of the NAM and GAM branches, and then applies CBAM attention to produce the final output. As shown in Fig. 7a above, the CBAM is a lightweight, versatile module. In the feature

maps, CBAM sequentially infers attention maps along two independent dimensions (channel and spatial), and then multiplies the attention maps with the input feature maps to perform adaptive feature optimization.

As shown in Fig. 7b, the Channel Attention Module compresses the feature map along the spatial dimensions to obtain a one-dimensional vector for further processing. During compression along the spatial dimensions, both average pooling and max pooling are considered. The Channel Attention Module can be expressed as follows:

$$\begin{aligned} M_c(F) &= \sigma(MLP(AvgPool(F)) + (MLP(MaxPool(F))) \\ &= \sigma(W_1(W_0)F^c_{avg})) + W_1(W_0(F^c_{\max}))) \end{aligned} \tag{7}$$

Similarly, as shown in Fig. 7c, the Spatial Attention Module compresses the channels of the feature map, performing average pooling and max pooling along the channel dimensions. The max pooling operation extracts the maximum values along the channels, repeated for the height multiplied by the width. The avg pooling operation extracts the average values along the channels, repeated for the height multiplied by the width. Then, the previously extracted feature maps (each with 1 channel) are combined to obtain a 2-channel feature map. The Spatial Attention Module can be expressed as follows:

$$\begin{aligned} M_c(F) &= \sigma\left(f^{7\times7}([AvgPool(F); MaxPool(F)])\right) \\ &= \sigma\left(f^{7\times7}\left[F^s_{\text{avg}}; F^s_{\max}\right]\right) \end{aligned} \tag{8}$$

where σ denotes the sigmoid operation, 7* 7 represents the size of the convolution kernel. A 7*7 convolution kernel has been found to be more effective compared to a 3*3 convolution kernel.

GAM attention branch (D)

The Derivative (D) branch extracts high-frequency features to predict boundary regions.

Figure 8 illustrates the detailed structure of GAM (D) Branch. The input of this branch is the output from the CBAM branch after two SWCA extractions, which has reached the deep stage of feature extraction. The GAM Branch itself undergoes one round of feature extraction, followed by a 1 × 3 CNN operation. Moreover, it employs six convolutional layers with kernel sizes of 5 × 5, 5 × 5, 5 × 5, 5 × 5, 4 × 4, and 3 × 3, respectively, for the coalesced extraction of multi-scale spatial features. Subsequently, the output of channel and spatial attention operations is concatenated with that of the CBAM branch. Finally, the combined features pass through the GAM module, which is most suitable for this branch, to produce the final output.

The GAM, as shown in Fig. 8a above, enhances network performance by reducing information diffusion and amplifying global interactive representations. It consists of both a channel attention module and a spatial attention module. As illustrated in Fig. 8b above. The channel attention submodule preserves information across three dimensions using a three-dimensional array. It then employs a two-layer MLP to amplify cross-dimensional channel-space dependencies. The MLP is an encoder-decoder structure similar to BAM, with a compression ratio of *r*. As shown in Fig. 8c above, the spatial attention submodule utilizes two convolutional layers to fuse spatial information, leveraging the same compression ratio r from the channel attention submodule. Furthermore, to maintain feature mappings, pooling operations are omitted due to their potential negative impact caused by

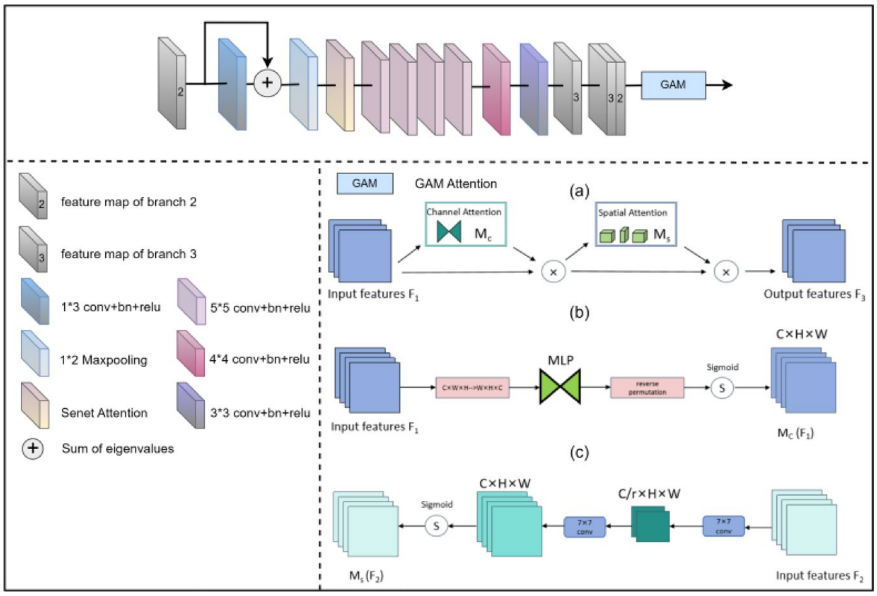


Fig. 8. Network structure of the GAM (D) branch.

reduced information utilization through max-pooling operations. Given input feature mapping $F_1 \in R^{C\times H\times W}$, intermediate states F_2 and F_3 are defined as:

$$F_2 = M_C(F_1) \otimes F_1 \tag{9}$$

$$F_3 = M_S(F_2) \otimes F_2 \tag{10}$$

F_1 represents the input. After passing F_1 through a three-dimensional arrangement and MLP operations, followed by a sigmoid activation function, we obtain $M_c(F_1)$. Then, it is multiplied by the original input F_1 to get F_2. F_2 is processed through a 7×7 convolutional layer, undergoes channel reduction, and then passes through another 7×7 convolution to obtain a new feature map. After applying an activation function, it results in $M_s(F_2)$. Finally, $M_s(F_2)$ is multiplied with the spatially obtained F_2 to yield the final input F_3.

Experimental setup and results

Data set

Similar to the approach in Ref.[52–54], this paper uses the Houston 2013, Salinas Valley (SV), and University of Pavia hyperspectral datasets to evaluate the proposed method. For each dataset, the values are first normalized to the range of 0–1. Then, for each class, 200 pixels along with their surrounding 5×5 neighborhoods are randomly selected as training samples, while the remaining pixels are used as test samples.

The Houston (HT) dataset has a data size of 349*1905, encompassing 144 spectral bands ranging from 364 to 1046nm. The ground truth labels include 15 land cover categories such as trees and soil, totaling 15,029 pixels. The selected categories and the number of samples are shown in Table 1.

The Salinas Valley (SV) dataset, after the removal of 20 water vapor and noise bands, retains 204 spectral bands. The dataset has a spatial dimension of 512×217 pixels with a spatial resolution of 3.7 m. The ground truth is categorized into 16 land cover classes, with specific land cover types and pixel counts detailed in Table 2.

The University of Pavia (UP) dataset, after the removal of noise and other bands, retains 103 spectral bands. The image size is 610×340 pixels, with a spatial resolution of 1.3 m. The spectral range spans from 0.43 to 0.86 μm. Approximately 20% of the pixels are labeled as ground truth, covering various urban structures, soil, natural targets, and shadows, among others. The specific pixel counts are shown in Table 3.

Parameter setting

The model parameters are set based on experience: the optimizer used is Adam, the learning rate is set to 0.00001, and the batch size for the IP and UP datasets is 256. The loss function is cross-entropy loss.

To evaluate the effectiveness of the proposed method in this paper, several approaches were employed, including support vector machines (SVM)[4], multinomial logistic regression (MLR)[5], extreme learning machines (ELM)[10], random forests (RF)[55], CNN[49], PPF[50], SpectralFormer (SF)[51], RAMiT[56], MSAA[57], SAFM[58], SSFTT[59], morphFormer[60] and GSC-ViT[61]. These methods were tested on the same training and test datasets. In the evaluation of the accuracy of different classification methods, overall accuracy (OA), average accuracy (AA), and Kappa coefficient are commonly employed.

The classification performance of the HT dataset

As shown in Fig. 9, the performance of various methods on the HT dataset is presented. It can be observed that traditional machine learning methods do not perform well, exhibiting numerous classification errors. All

No	ClassName	Training Num	Testing Num	All Num
0	Background	–	–	5712
1	grass_healthy	200	1051	1251
2	grass_stressed	200	1054	1254
3	grass_synergic	200	497	697
4	tree	200	1044	1244
5	soil	200	1042	1242
6	water	200	125	325
7	residential	200	1068	1268
8	commercial	200	1044	1244
9	road	200	1052	1252
10	highway	200	1027	1227
11	railway	200	1035	1235
12	parking lot 1	200	1033	1233
13	parking lot 2	200	269	469
14	tennis court	200	228	428
15	running track	200	460	660
	Total	3000	12,029	20,741

Table 1. Sample selection for the Houston dataset.

No	ClassName	Training Num	Testing Num	All Num
0	Background	–	–	56,975
1	Brocoli-gree-weeds-1	200	1809	2009
2	Brocoli-gree-weeds-2	200	3526	3726
3	Fallow	200	1776	1976
4	Fallow-rough-plow	200	1194	1394
5	Fallow-smooth	200	2478	2678
6	Stubble	200	3759	3959
7	Celery	200	3379	3579
8	Grapes-untrained	200	11,071	11,271
9	Soil-vinyard-develop	200	6003	6203
10	Corn-senesced-green-weeds	200	3078	3278
11	Lettuce-romaine-4wk	200	868	1068
12	Lettuce-romaine-5wk	200	1727	1927
13	Lettuce-romaine-6wk	200	716	916
14	Lettuce-romaine-7wk	200	870	1070
15	Vinyard-untrained	200	7068	7268
16	Vinyard-vertical-trellis	200	1607	1807
	Total	3200	50,929	111,104

Table 2. Sample selection for the Salinas Valley dataset.

No	ClassName	Training Num	Testing Num	All Num
0	Background	–	–	164,624
1	Asphalt	200	6431	6631
2	Meadows	200	18,449	18,649
3	Gravel	200	1899	2099
4	Trees	200	2864	3064
5	Painted metal sheets	200	1145	1345
6	Bare Soil	200	4829	5029
7	Bitumen	200	1130	1330
8	Self-Blocking Bricks	200	3482	3682
9	Shadows	200	747	947
	Total	1800	40,976	207,400

Table 3. Sample selection for the University of Pavia dataset.

methods show relatively good performance on the "grass_synergic" category. Deep learning-based methods perform well overall, with both SAFM and the proposed method demonstrating strong performance across all categories. As shown in the enlarged area in the figure, apart from the morphFormer and GSC-ViT methods, other methods have a relatively large number of misclassifications. The classification result of the proposed method in this paper is completely correct in this area. Notably, the method proposed in this paper stands out even more, demonstrating markedly superior classification performance.

As shown in Table 4, the classification accuracy of each method for every category, as well as the overall accuracy (OA), average accuracy (AA), and Kappa coefficient, are presented. From the table, it can be seen that all methods achieve an accuracy exceeding 98% for the "grass_synergic" category. However, for other categories, MLR performs significantly worse than the other methods, resulting in the lowest OA and Kappa coefficient. SF and SFAM show better performance, while the proposed method performs well across all categories, with an OA that is 1.09% higher than the second-ranked SAFM and nearly 40% higher than the worst-performing MLR. In the tennis court class, all methods demonstrate satisfactory classification performance. The classification accuracy of the proposed method in this paper, as well as SSFTT, morphFormer, and GSC-ViT which are specifically designed for hyperspectral image classification, reaches 100%. The results in the table demonstrate that the method proposed in this paper exhibits noticeably superior classification performance.

The classification performance of the SV dataset

As shown in Fig. 10, the classification performance of various methods on the SV dataset is presented. It can be observed that traditional machine learning methods do not perform well, exhibiting numerous classification errors. However, all methods perform well on the "Lettuce-romaine-5wk" category. Several deep learning-based methods show excellent performance on the "Vinyard-vertical-trellis" category. Both SF and SAFM demonstrate

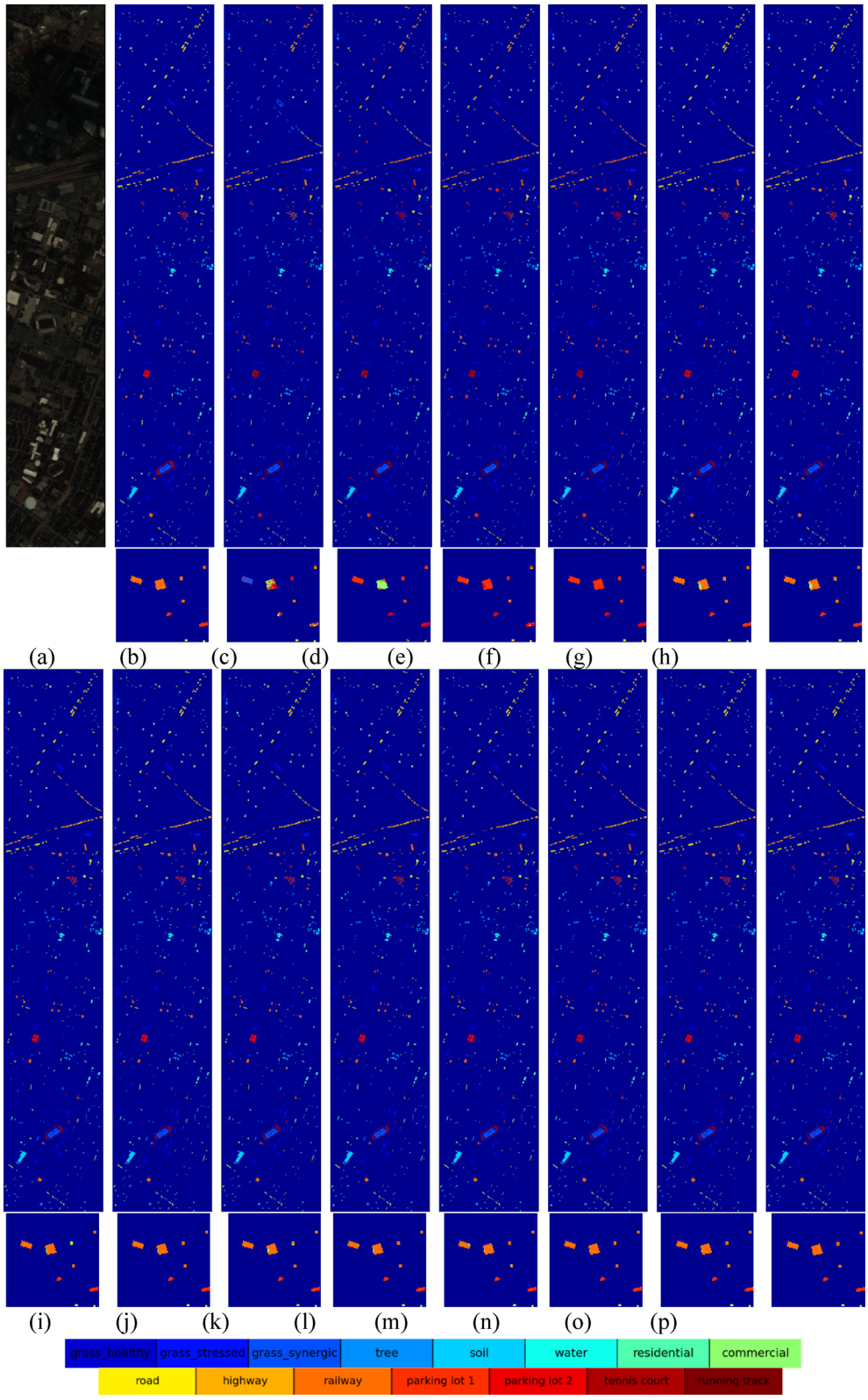


Fig. 9. The classification performance of various methods on the HT dataset: (**a**) Original HSI (**b**) Ground Truth Map (**c**) MLR (**d**) SVM (**e**) RF (**f**) ELM (**g**) CNN2D (**h**) PPF (**i**) SF (**j**) RAMiT (**k**) MSAA (**l**) SAFM (**m**) SSFTT (**n**) morphFormer (**o**) GSC-ViT (**p**) PMMFNet.

	MLR	SVM	RF	ELM	CNN	PPF	SF	RAMiT	MSAA	SAFM	SSFTT	morphFormer	GSC-ViT	PMMF
grass_healthy	86.39	98.57	98.19	86.11	100	99.71	99.14	97.91	96.76	92.77	100	100	100	100
grass_stressed	76.66	88.99	96.87	97.91	99.05	99.34	99.62	99.53	99.24	99.72	99.53	99.53	99.72	99.72
grass_synergic	98.79	99.8	97.38	100	99.8	99.6	100	98.99	99.6	100	99.8	99.8	99.8	99.8
tree	94.73	92.72	99.04	98.85	95.69	100	98.85	99.62	99.81	98.28	100	100	100	100
soil	98.66	99.14	98.37	99.81	99.9	99.81	100	98.75	99.81	99.81	100	100	99.9	100
water	88.8	90.4	96.8	100	98.4	100	100	97.6	100	100	100	100	100	100
residential	54.78	80.34	84.93	91.29	91.29	95.6	97.85	87.36	88.67	98.22	98.31	93.73	97	97.66
commercial	50.77	42.82	80.27	57.76	94.83	95.31	97.32	82.85	91.09	98.37	97.99	98.47	99.62	99.23
road	35.46	73.48	78.71	69.49	87.26	90.59	95.72	79.75	85.74	97.72	96.96	95.72	97.34	97.91
highway	22.01	79.36	82.57	63.68	98.64	95.03	96.79	91.43	93.09	99.12	99.42	99.51	99.42	99.51
railway	28.89	62.32	78.36	57.2	92.95	91.11	91.79	84.06	89.76	99.23	96.62	97.29	99.03	98.94
parking lot 1	9.97	57.6	74.93	54.5	96.71	93.22	96.03	89.55	82.38	96.13	97.1	99.42	97	98.74
parking lot 2	10.04	36.06	52.42	74.35	92.94	99.26	99.26	94.42	98.51	99.63	99.63	99.26	99.63	99.63
tennis court	86.84	98.25	99.12	99.56	97.37	100	99.12	99.12	98.68	98.68	100	100	100	100
running track	99.78	99.35	99.13	98.91	100	100	100	100	100	100	100	100	100	100
OA	59.31	78.93	87.67	80.06	95.96	96.48	97.64	92.04	93.53	98.17	98.75	98.55	99.02	99.26
AA	62.84	79.95	87.81	83.29	96.32	97.24	98.1	93.4	94.88	98.51	99.02	98.85	99.23	99.41
Kappa	0.561	0.771	0.866	0.783	0.956	0.961	0.974	0.913	0.929	0.980	0.986	0.9842	0.9894	0.992

Table 4. Classification performance of various methods on the HT dataset.

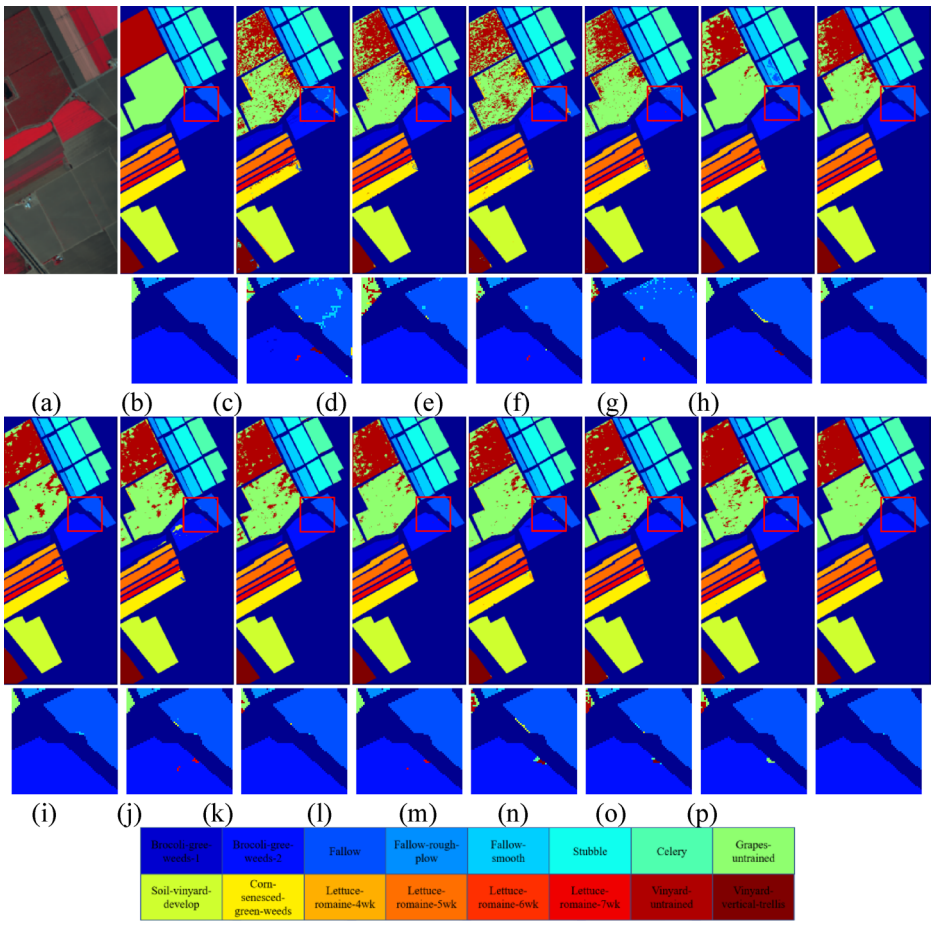


Fig. 10. The classification performance of various methods on the SV dataset: (**a**) Original HSI (**b**) Ground Truth Map (**c**) MLR (**d**) SVM (**e**) RF (**f**) ELM (g) CNN2D (**h**) PPF (**i**) SF (**j**) RAMiT (**k**) MSAA (**l**) SAFM (**m**) SSFTT (**n**) morphFormer (**o**) GSC-ViT (**p**) PMMFNet.

relatively good overall classification capabilities, SSFTT and morphFormer perform exceptionally well on the Vinyard-untrained class. As shown in the magnified area of the figure, compared with other methods, the method proposed in this paper demonstrates the best classification results. The method proposed in this paper stands out with superior performance, offering better classification capabilities.

As shown in Table 5, the classification accuracy of each method is highly distinctive. Among them, the accuracy of each method in the Fallow-rough-plow category is close to 100%. However, in other categories, the performance of MLR and RF is significantly lower than that of other methods, resulting in the lowest accuracy. The accuracy of SF, RAMiT, MSAA, and ELM is relatively similar, with SAFN performing slightly better. In the Vinyard-untrained class, the methods proposed in recent years have all performed quite well, with classification accuracies close to or exceeding 90%. However, the GSC-ViT method only has a classification accuracy of 80.31%. The method proposed in this paper performs excellently in all categories, with noticeably higher values in OA, AA, and Kappa coefficient compared to other methods. Based on the results in the table, it can be concluded that the proposed method has superior classification performance.

The classification performance of the UP dataset

Figure 11 shows the classification performance of various methods on the UP dataset. It can be observed that in the categories with the largest number of pixels, such as Asphalt, Bare Soil, and Meadows, the classification errors of each method reflect their classification performance. MLR exhibits the most classification errors, followed by RF and ELM. SF, CNN, and RAMiT demonstrate good classification performance, while PPF, MSAA, and SAFM perform very well in terms of misclassification. The classification results of SSFTT and mornpFormer are satisfactory. However, there are numerous misclassifications on the Self-Blocking Bricks class. As shown in the magnified area of the figure, other methods exhibit more misclassifications, while the method presented in this paper performs well.The method proposed in this paper demonstrates the overall best classification performance.

As shown in Table 6 below, overall, the MLR method has the lowest OA, while SVM performs the best among the four traditional machine learning algorithms, though it is slightly inferior to deep learning-based methods. The classification accuracies of PPF and SAFM reach 97.87% and 98.73%, respectively, demonstrating excellent classification performance, while the other methods show relatively lower accuracy. The classification result of GSC-ViT on the Gravel class is only 96.54, which affects its overall classification performance The proposed method in this paper achieves an overall accuracy of up to 99.65%, indicating that it has the highest classification performance on this dataset.

Classification performance with fewer samples

To compare the performance of various methods under the influence of fewer training samples, this paper sets the number of training samples per class to 50, 100, and 200 for training, following the same rationale. The classification results of each method are computed, as shown in the subsequent Fig. 12. It is clearly evident that as the sample size increases, there is a significant upward trend in the classification performance across all datasets, aligning with our expectations. Traditional machine learning algorithms like SVM maintain relatively strong classification performance due to their stable behavior. However, ELM's performance is less stable on these three datasets. This is because the weight matrix from input neurons to hidden neurons and the hidden layer thresholds are randomly generated, which can lead to ill-conditioned output matrices and unstable network

	MLR	SVM	RF	ELM	CNN	PPF	SF	RAMiT	MSAA	SAFM	SSFTT	morphFormer	GSC-ViT	PMMF
Brocoli-gree-weeds-1	97.82	99.31	99.48	99.71	100	100	98.85	97.99	98.45	99.48	99.94	100	99.94	100
Brocoli-gree-weeds-2	96.85	99.94	99.49	99.80	99.77	100	100	95.77	99.97	99.83	99.21	100	99.89	100
Fallow	92.17	99.49	97.41	97.69	99.38	99.77	99.61	99.66	99.89	100	99.89	99.83	100	99.83
Fallow-rough-plow	99.25	99.75	99.75	99.75	99.92	99.92	100	100	100	99.92	99.91	100	100	99.92
Fallow-smooth	97.50	98.14	96.17	98.55	86.88	97.74	99.35	98.95	98.55	99.11	98.51	99.11	97.54	99.39
Stubble	99.36	99.73	99.76	99.57	99.89	99.89	100	100	100	99.95	99.95	99.65	100	99.97
Celery	99.34	99.88	98.78	99.70	99.79	99.85	99.97	99.91	98.06	99.73	99.94	99.7	99.28	99.94
Grapes-untrained	74.01	86.69	73.49	85.86	98.19	89.67	89.81	87.92	86.07	94.4	82.89	84.05	97.78	97.18
Soil-vinyard-develop	96.58	99.97	98.93	99.95	100	100	99.87	100	99.97	100	100	100	100	99.98
Corn-senesced-green-weeds	85.57	95.96	91.29	94.84	95.70	98.48	98.38	97.52	99.77	99.74	99.83	99.93	98.68	99.34
Lettuce-romaine-4wk	92.68	99.29	97.99	96.93	97.87	99.65	100	99.65	99.65	99.29	99.88	100	94.68	99.65
Lettuce-romaine-5wk	99.64	99.94	98.28	99.29	100	100	100	100	100	100	100	100	100	100
Lettuce-romaine-6wk	99.14	99.71	96.72	99.14	99.86	100	100	99.86	100	99.86	100	100	100	100.00
Lettuce-romaine-7wk	90.95	98.94	94.71	94.36	99.41	99.76	99.76	99.53	96	99.65	95.92	99.52	99.76	99.88
Vinyard-untrained	65.09	68.66	65.56	67.03	70.31	84.23	88.54	87.63	91.24	88.31	99.53	99.76	80.31	95.02
Vinyard-vertical-trellis	92.12	99.38	98.41	99.24	100	99.59	99.93	100	100	100	100	100	100	100
OA	86.94	92.34	88.13	91.65	94.53	95.33	96.00	95.07	95.41	97.04	96.09	96.38	96.47	98.61
AA	92.38	96.55	94.14	95.71	96.69	98.03	98.38	97.77	97.98	98.7	98.46	98.85	97.99	99.38
Kappa	0.854	0.914	0.867	0.906	0.938	0.947	0.955	0.945	0.949	0.967	0.954	0.959	0.961	0.984

Table 5. Classification performance of various methods on the SV dataset.

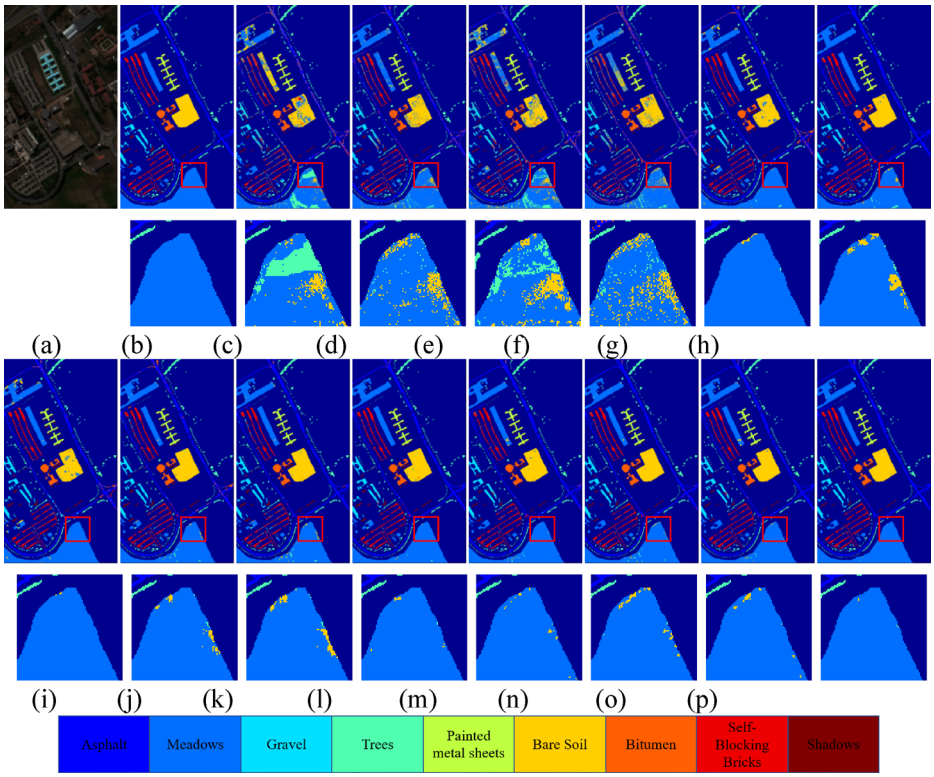


Fig. 11. The classification performance of various methods on the UP dataset: (**a**) Original HSI (**b**) Ground Truth Map (**c**) MLR (**d**) SVM (**e**) RF (**f**) ELM (**g**) CNN2D (**h**) PPF (**i**) SF (j) RAMiT (**k**) MSAA (**l**) SAFM (**m**) SSFTT (**n**) morphFormer (**o**) GSC-ViT (**p**) PMMFNet.

	MLR	SVM	RF	ELM	CNN	PPF	SF	RAMiT	MSAA	SAFM	SSFTT	morphFormer	GSC-ViT	PMMF
Asphalt	73.11	86.67	78.14	64.08	85.83	97.79	89.20	89.2	97.72	98.31	97.26	97.86	98.86	99.48
Meadows	73.23	93.10	75.84	87.51	96.96	97.95	96.72	98.15	97.5	99.29	98.86	98.37	98.74	99.73
Gravel	75.65	83.98	80.30	82.52	92.91	90.69	97.56	99.13	99.84	95.83	97.46	97.62	96.54	99.51
Trees	95.70	97.60	96.94	96.37	99.47	97.92	98.13	99.54	99.68	98.63	99.58	99.72	99.47	99.82
Painted metal sheets	98.52	98.60	98.52	99.83	100	100	100	99.91	99.83	100	100	100	100	100
Bare soil	76.21	88.88	81.03	87.51	93.46	99.19	93.60	99.36	99.19	98.53	99.34	99.69	99.65	99.73
Bitumen	90.97	93.98	90.88	94.96	95.49	98.85	93.81	100	99.29	98.94	97.88	98.5	98.5	100
Self-blocking bricks	79.12	86.01	81.79	68.72	95.98	98.10	94.23	94.34	90.58	97.82	94.92	96.09	97.21	99.02
Shadows	100	100	100	100	100	99.87	100	100	100	100	100	100	100	100
OA	77.51	91.17	80.57	83.37	94.79	97.87	95.15	96.83	97.57	98.73	98.34	98.39	98.74	99.65
AA	84.72	92.09	87.05	86.83	95.57	97.82	95.92	97.74	98.18	98.59	98.37	98.65	98.77	99.70
Kappa	0.713	0.883	0.751	0.783	0.931	0.971	0.935	0.958	0.968	0.983	97.78	97.86	98.32	0.995

Table 6. Classification performance of various methods on the UP dataset.

structures, reducing robustness and classification performance. Deep learning-based methods noticeably outperform traditional machine learning methods across all training sample sizes on the three datasets. The classification accuracy of Methods SSFTT, morphFormer, and GSC-ViT is relatively similar, while the proposed method consistently demonstrates the best classification performance.

The impact of different sample sizes

Due to the limited number of samples in hyperspectral images, this paper adopts a method of random generation to create one hundred thousand samples within each class, aiming to increase the sample size and enhance classification performance. Generally speaking, the more samples there are, the better the learning and extraction of more detailed features can be achieved, but this also significantly increases computational load. To evaluate the impact of sample size on classification performance, this paper sets four levels of sample sizes: 10,000, 50,000, 100,000, and 200,000. Using the network structure proposed in this paper, the final classification performance is assessed. The results, as shown in Fig. 13, indicate that as the sample size increases, the network's classification

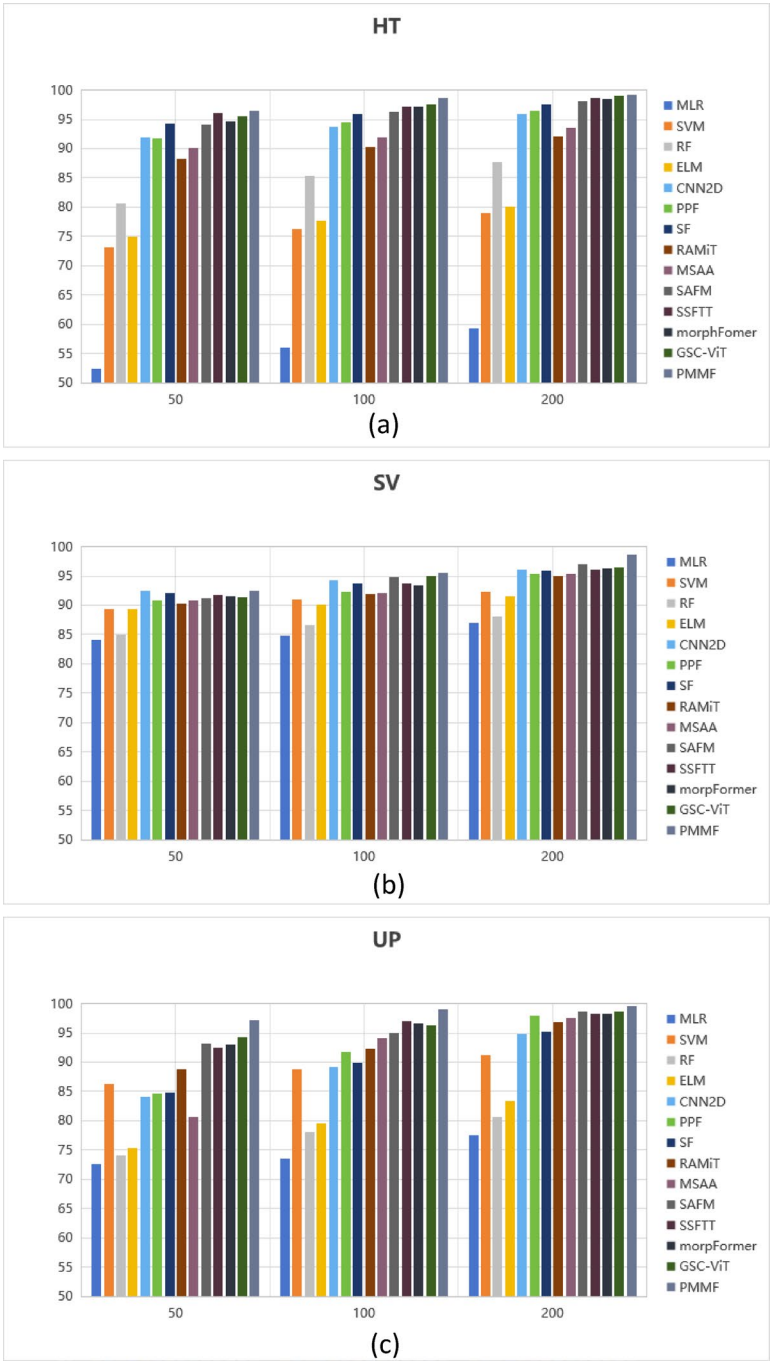


Fig. 12. Classification performance of various methods with fewer samples on (**a**) HT, (**b**) SV, and (**c**) UP.

performance improves. When the sample size reaches 100,000, the classification accuracy tends to plateau. To balance the classification and computing performance, the number of test samples in this paper is set at 100,000.

Classification performance of different attention branches

This paper employs three different attention branches, encompassing various sizes and scales, to compare the performance of three distinct attention mechanisms—NAM, GAM, and CBAM—across these branches. A total of six different combination structures were considered. Using the same methodology, the classification performance of each combination structure was calculated, with a test sample size of 100,000. The results, as described in Fig. 14, indicate that the use of different attention mechanisms has a certain impact on the classification performance due to the varying dimensions of the feature maps processed by each branch. Comparing the OA, AA, and Kappa coefficients of the six different combination structures, it is evident that the structure adopted in this paper, with Branch 1 using NAM, Branch 2 using CBAM, and Branch 3 using GAM,

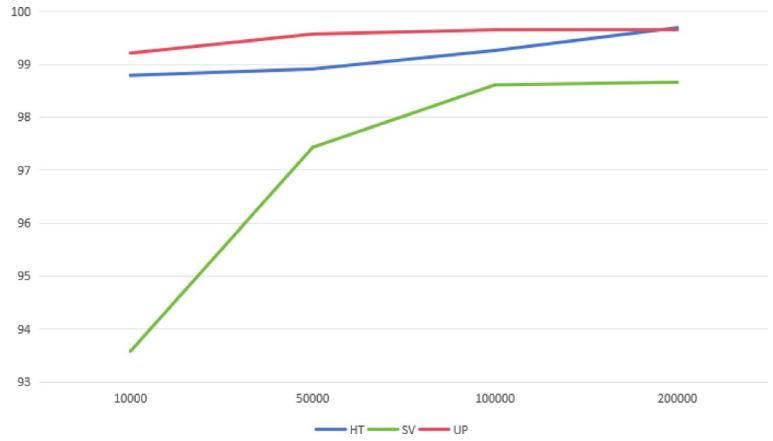


Fig. 13. Classification performance of the proposed method under different numbers of samples.

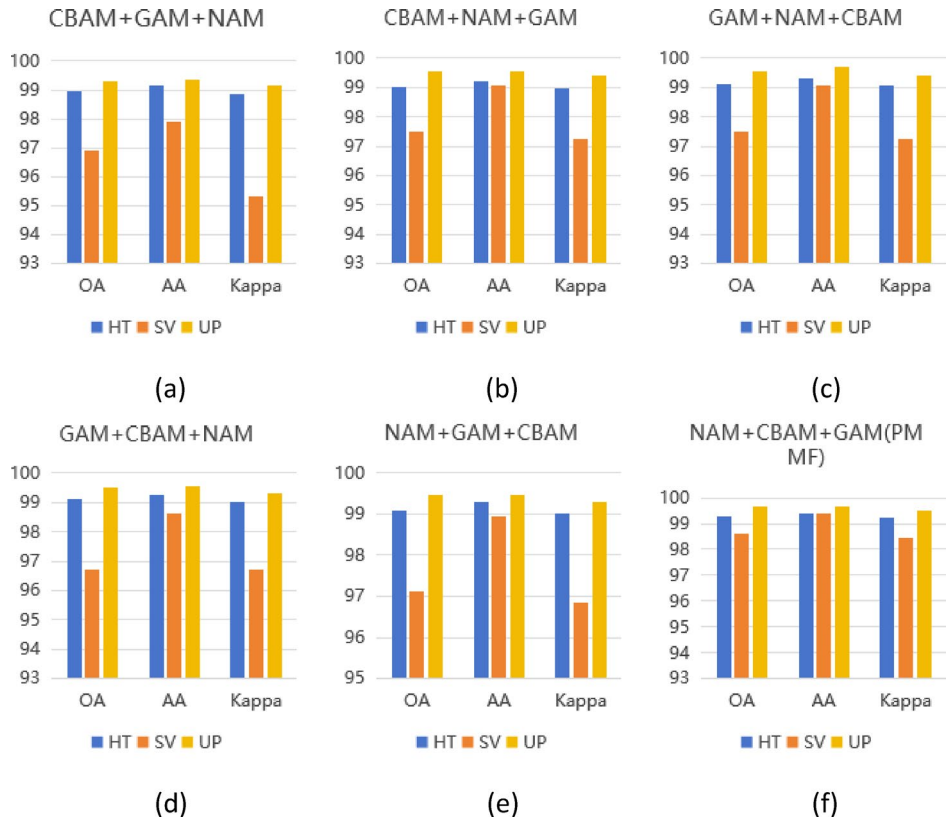


Fig. 14. The performance of different attention mechanisms in different branchs of the PMMFNet method. (**a**) CBAM + GAM + NAM (**b**) CBAM + NAM + GAM (**c**) GAM + CBAM + NAM (**d**) GAM + NAM + CBAM (**e**) NAM + GAM + CBAM (**f**) NAM + CBAM + GAM (in this paper).

is noticeably more stable than the other five combinations. Moreover, this structure also achieved much higher classification accuracy across three datasets compared to the other five.

Classification performance of shuffle or no-shuffle

Because of the small sample size of hyperspectral datasets, there is a risk of overfitting when using few-shot learning methods with deep learning models. Consequently, effectively expanding the sample size is a critical issue. This paper employs a spatial structure based on neighboring pixel elements and proposes a spatial shuffle scheme for small patches. On this foundation, using the network architecture presented in this paper, a relatively high classification accuracy is achieved. To compare the impact of shuffle versus no-shuffle on classification outcomes, this paper conducts tests under the condition of selecting 100,000 samples and training with 200

	Shuffle	No-shuffle
Houston	99.26	97.83
Salinas Valley	98.61	94.17
University of Pavia	99.65	97.96

Table 7. Shuffle or no-Shuffle on various datasets.

patch sizes	Houston	Salinas Valley (SV)	University of Pavia (UP)
5×5	99.26	98.61	99.65
7×7	99.21	98.69	99.47
9×9	99.22	98.79	98.62

Table 8. Classification performance on different datasets corresponding to various patch sizes.

samples. The data in the Table 7 below clearly indicates that the classification performance obtained after shuffle processing is noticeably superior to the no-shuffle classification performance.

Classification performance of different patch sizes

To assess the effect of spatial context granularity on classification performance, we designed comparative experiments with different patch sizes (5×5, 7×7, 9×9 pixels) under the condition of selecting 100,000 samples and training with 200 samples. The results from the Table 8 below show that on the Houston dataset, there is little difference in classification results among the three different patch sizes. On the SV dataset, classification performance slightly improves as the size increases. On the UP dataset, increasing the size introduces irrelevant noise, leading to a decrease in classification accuracy. Therefore, this paper selects a 5×5 patch size to balance accuracy and computational efficiency, achieving optimal results.

Conclusion

In this paper, we propose a Progressive Multi-Scale Multi-Attention Fusion Network, referred to as PMMFNet. Inspired by the PID controller, PMMFNet progressively extracts feature information from multiple scales through three complementary branches: Proportional (P), Integral (I), and Derivative (D). SWCA module is designed to improve the model's ability to extract prominent features using skip connection-based feature fusion. Additionally, a multi-attention branch fusion module is proposed, which stacks features from different scales and selects the most appropriate attention mechanism for feature absorption, effectively enhancing feature reusability. Experimental results show that PMMFNet achieves high accuracy. Future work will focus on reducing the sample size while improving classification accuracy.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Received: 4 March 2025; Accepted: 4 August 2025
Published online: 11 August 2025

References

1. Ding, Y. et al. Unsupervised self-correlated learning smoothy enhanced locality preserving graph convolution embedding clustering for hyperspectral images. *IEEE Trans. Geosci. Remote Sens.* **60**, 1–16. https://doi.org/10.1109/TGRS.2022.3202865 (2022).
2. Yang, X. & Yu, Y. Estimating soil salinity under various moisture conditions: An experimental study. *IEEE Trans. Geosci. Remote Sens.* **55**(5), 2525–2533. https://doi.org/10.1109/TGRS.2016.2646420 (2017).
3. Yokoya, N., Chan, J.C.-W. & Segl, K. Potential of resolution enhanced hyperspectral data for mineral mapping using simulated EnMAP and sentinel-2 images. *Remote Sens.* **8**(3), 172–189. https://doi.org/10.3390/rs8030172 (2016).
4. Li, S., Dian, R., Fang, L. & Bioucas-Dias, J. M. Fusing hyperspectral and multispectral images via coupled sparse tensor factorization. *IEEE Trans. Image Process.* **27**(8), 4118–4130. https://doi.org/10.1109/TIP.2018.2836307 (2018).
5. Zhang, S., Li, J., Wu, Z. & Plaza, A. Spatial discontinuity-weighted sparse unmixing of hyperspectral images. *IEEE Trans. Geosci. Remote Sens.* **56**(10), 5767–5779. https://doi.org/10.1109/TGRS.2018.2825457 (2018).
6. Kumar, V. & Ghosh, J. K. Camouflage detection using MWIR hyperspectral images. *J Indian Soc. Remote Sens.* **45**, 139–145. https://doi.org/10.1007/s12524-016-0555-8 (2017).
7. Zhang, Q., Yuan, Q., Song, M., Yu, H. & Zhang, L. Co-operated spectral low-rankness prior and deep spatial prior for HSI unsupervised denoising. *IEEE Trans. Image Process.* **31**, 6356–6368. https://doi.org/10.1109/TIP.2022.3211471 (2022).
8. Ding, Y., Zhang, Z., Kang, W., Yang, A., Zhao, J., Feng, J., Hong, D. & Zheng, Q. Adaptive homophily clustering: A structure homophily graph learning with adaptive filter for hyperspectral image. Preprint at arXiv:2501.01595 (2025).
9. Heiden, U., Roessner, S., Segl, K. & Kaufmann, H. Analysis of spectral signatures of urban surfaces for their identification using hyperspectral HyMap data, in *IEEE/ISPRS Joint Workshop on Remote Sensing and Data Fusion over Urban Areas (Cat. No.01EX482)*, Rome, Italy, 173–177, https://doi.org/10.1109/DFUA.2001.985871 (2001).
10. Melgani, F. & Bruzzone, L. Classification of hyperspectral remote sensing images with support vector machines. *IEEE Trans. Geosci. Remote Sens.* **42**(8), 1778–1790. https://doi.org/10.1109/TGRS.2004.831865 (2004).

11. Li, J., Bioucas-Dias, J. M. & Plaza, A. Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning. *IEEE Trans. Geosci. Remote Sens.* **48**(11), 4085–4098. https://doi.org/10.1109/TGRS.2010.2060550 (2010).
12. Liu, Z., Tang, B., He, X., Qiu, Q. & Liu, F. Class-specific random forest with cross-correlation constraints for spectral—spatial hyperspectral image classification. *IEEE Geosci. Remote Sens. Lett.* **14**(2), 257–261. https://doi.org/10.1109/LGRS.2016.2637561 (2017).
13. Du, B. & Zhang, L. Target detection based on a dynamic subspace. *Pattern Recognit.* **47**(1), 344–358. https://doi.org/10.1016/j.patcog.2013.07.005 (2014).
14. Yu, H. et al. Hyperspectral image classification based on adjacent constraint representation. *IEEE Geosci. Remote Sens. Lett.* **18**(4), 707–711. https://doi.org/10.1109/LGRS.2020.2982706 (2021).
15. Le, L., Xie, Y. & Raghavan, V. V. KNN loss and deep KNN. *Fund. Inf.* **182**(2), 95–110. https://doi.org/10.3233/FI-2021-2068 (2021).
16. Villa, A., Benediktsson, J. A., Chanussot, J. & Jutten, C. Hyperspectral image classification with independent component discriminant analysis. *IEEE Trans. Geosci. Remote Sens.* **49**(12), 4865–4876. https://doi.org/10.1109/TGRS.2011.2153861 (2011).
17. Licciardi, G., Marpu, P. R., Chanussot, J. & Bened-iktsson, J. A. Linear versus nonlinear PCA for the classification of hyperspectral data based on the extended morphological profiles. *IEEE Geosci. Remote Sens. Lett.* **9**(3), 447–451. https://doi.org/10.1109/LGRS.2011.2172185 (2012).
18. Bandos, T. V., Bruzzone, L. & Camps-Valls, G. Classification of hyperspectral images with regularized linear discriminant analysis. *IEEE Trans. Geosci. Remote Sens.* **47**(3), 862–873. https://doi.org/10.1109/TGRS.2008.2005729 (2009).
19. Ghamisi, P. et al. New frontiers in spectral-spatial hyperspectral image classification: The latest advances based on mathematical morphology, Markov random fields, segmentation, sparse representation, and deep learning. *IEEE Geosci. Remote Sens. Mag.* **6**(3), 10–43. https://doi.org/10.1109/MGRS.2018.2854840 (2018).
20. Feng, J., Zhang, T., Zhang, J., Shang, R., Dong, W., Shi, G. & Jiao, L. S 4 DL: Shift-sensitive spatial–spectral disentangling learning for hyperspectral image unsupervised domain adaptation. *IEEE Transactions on Neural Networks and Learning Systems*. Preprint at arXiv:2408.15263 (2025).
21. Khurshid, H. & Khan, M. F. Classification of remotely sensed images using decimal coded morphological profiles. *SIViP* **10**, 1001–1007. https://doi.org/10.1007/s11760-015-0851-8 (2016).
22. Benediktsson, J. A., Palmason, J. A. & Sveinsson, J. R. Classification of hyperspectral data from urban areas based on extended morphological profiles. *IEEE Trans. Geosci. Remote Sens.* **43**(3), 480–491. https://doi.org/10.1109/TGRS.2004.842478 (2005).
23. Liu, S., Zhang, L., Cen, Y., Chen, L. & Wang, Y. A fast hyperspectral anomaly detection algorithm based on greedy bilateral smoothing and extended multi-attribute profile. *Remote Sens.* **13**, 3954. https://doi.org/10.3390/rs13193954 (2021).
24. Ding, Y., Zhang, Z., Yang, A., Cai, Y., Xiao, X., Hong, D. & Yuan, J., SLCGC: A lightweight self-supervised low-pass contrastive graph clustering network for hyperspectral images. Preprint at arXiv:2502.03497 (2025).
25. Bordes, A., Glorot, X., Weston, J. & Bengio, Y. Joint learning of words and meaning representations for open-text semantic parsing. in *Proc.Int. Conf. Art. Intell. Stat*, pp. 127–135 (2012).
26. Zheng, W. -L., Zhu, J. -Y., Peng, Y. & Lu, B. -L. EEG-based emotion classification using deep belief networks. in *2014IEEE International Conference on Multimedia and Expo (ICME)*, Chengdu, China, pp. 1–6, https://doi.org/10.1109/ICME.2014.6890166 (2014)
27. Lin, Z., Chen, Y., Zhao, X. & Wang, G. Spectral-spatial classification of hyperspectral image using autoencoders. in *2013 9th International Conference on Information, Communications & Signal Processing, Tainan*, pp. 1–5, https://doi.org/10.1109/ICICS.2013.6782778. (2013).
28. Li, D., Zhang, J., Zhang, Q. & Wei, X. Classification of ECG signals based on 1D convolution neural network. in *2017 IEEE 19th International Conference on e-Health Networking, Applications and Services (Healthcom)*, Dalian, China, pp. 1–6, https://doi.org/10.1109/HealthCom.2017.8210784 (2017)
29. Wang, Q., Yuan, Z., Du, Q. & Li, X. GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection. *IEEE Trans. Geosci. Remote Sens.* **57**(1), 3–13. https://doi.org/10.1109/TGRS.2018.2849692 (2019).
30. Li, Y., Zhang, H. & Shen, Q. Spectral–spatial classification of hyperspectral imagery with 3d convolutional neural network. *Remote Sens.* **9**(1), 67. https://doi.org/10.1016/j.asr.2019.05.005 (2019).
31. Roy, S. K., Krishna, G., Dubey, S. R. & Chaudhuri, B. B. HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification. *IEEE Geosci. Remote Sens. Lett.* **17**(2), 277–281. https://doi.org/10.1109/LGRS.2019.2918719 (2020).
32. Roy, S. K., Manna, S., Song, T. & Bruzzone, L. Attention—Based adaptive spectral-spatial Kernel ResNet for hyperspectral image classification. *IEEE Trans. Geosci. Remote Sens.* **59**(9), 7831–7843 (2021).
33. Lei, X., Pan, H. & Huang, X. A dilated CNN model for image classification. *IEEE Access* **7**, 124087–124095. https://doi.org/10.1109/ACCESS.2019.2927169 (2019).
34. Huang, G., Liu, Z., van der Maaten, L. & Weinberger, K. Q. Densely connected convolutional networks. in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 2261–2269 (2017).
35. Zhang, W., Tang, P. & Zhao, L. Remote sensing image scene classification using CNN-CapsNet. *Remote Sens.* **11**, 494. https://doi.org/10.3390/rs11050494 (2019).
36. Xu, F., Zhang, G., Song, C., Wang, H. & Mei, S. Multiscale and cross-level attention learning for hyperspectral image classification. *IEEE Trans. Geosci. Remote Sens.* **61**, 1–15. https://doi.org/10.1109/TGRS.2023.3235819 (2023).
37. Qiao, X., Roy, S. K. & Huang, W. Multiscale neighborhood attention transformer with optimized spatial pattern for hyperspectral image classification. *IEEE Trans. Geosci. Remote Sens.* **61**, 1–15. https://doi.org/10.1109/TGRS.2023.3314550 (2023).
38. Hu, B., Tu, Q., Ren, X., Liao, Z. C. & Plaza, A. Hyperspectral Image classification via multiscale multiangle attention network. *IEEE Trans. Geosci. Remote Sens.* **62**, 1–18. https://doi.org/10.1109/TGRS.2024.3370919 (2024).
39. Vaswani, A. et al., Attention is all you need. in *Proc. Adv. Neural Inf. Process. Syst.*, pp. 5998–6008 (2017).
40. Hu, J., Shen, L. & Sun, G. Squeeze-and-excitation networks. in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, pp. 7132–7141 (2018).
41. Wang, Q., Wu, B., Zhu, P., Li, P., Zuo, W. & Hu, Q. ECA-Net: Efficient channel attention for deep convolutional neural networks. in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 11534–11542 (2020).
42. Fu, J. et al. Dual attention network for scene segmentation. in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pp. 3146–3154 (2019).
43. Ma, W., Yang, Q., Wu, Y., Zhao, W. & Zhang, X. Double-branch multi-attention mechanism network for hyperspectral image classification. *Remote Sens.* **11**, 1307. https://doi.org/10.3390/rs11111307 (2019).
44. Li, R., Zheng, S., Duan, C., Yang, Y. & Wang, X. Classification of hyperspectral image based on double-branch dual-attention mechanism network. *Remote Sens.* **12**, 582. https://doi.org/10.3390/rs12030582 (2020).
45. Cui, Y., Yu, Z., Han, J., Gao, S. & Wang, L. Dual-triple attention network for hyperspectral image classification using limited training samples. *IEEE Geosci. Remote Sens. Lett.* **19**, 1–5. https://doi.org/10.1109/LGRS.2021.3067348 (2022).
46. Tang, Z. et al. Joint graph attention and asymmetric convolutional neural network for deep image compression. *IEEE Trans. Circuits Syst. Video Technol.* **33**(1), 421–433. https://doi.org/10.1109/TCSVT.2022.3199472 (2023).
47. Liu, T. et al. Spatial channel attention for deep convolutional neural networks. *Mathematics* **10**, 1750. https://doi.org/10.3390/math10101750 (2022).
48. Wang, Z., Cao, B. & Liu, J. Hyperspectral image classification via spatial shuffle-based convolutional neural network. *Remote Sens.* **15**, 3960. https://doi.org/10.3390/rs15163960 (2023).

49. Ham, J., Chen, Y., Crawford, M. M. & Ghosh, J. Investigation of the random forest framework for classification of hyperspectral data. *IEEE Trans. Geosci. Remote Sens.* **43**(3), 492–501. https://doi.org/10.1109/TGRS.2004.842481 (2005).
50. Paoletti, M. E., Haut, J. M., Plaza, J. & Plaza, A. Deep learning classifers for hyperspectral imaging: A review. *ISPRS J. Photogram. Remote Sens.* https://doi.org/10.1016/j.isprsjprs.2019.09.006 (2019).
51. Hong, D. et al. SpectralFormer: Rethinking hyperspectral image classification with transformers. *IEEE Trans. Geosci. Remote Sens.* **60**, 1–15. https://doi.org/10.1109/TGRS.2021.3130716 (2022).
52. Ding, H. et al. TBSSF-Net: Three-branch spatial-spectral fusion network for hyperspectral image classification. *Opt. Express* https://doi.org/10.1364/OE.550150 (2025).
53. Li, W., Wu, G., Zhang, F. & Du, Q. Hyperspectral image classification using deep pixel-pair features. *IEEE Trans. Geosci. Remote Sens.* **55**(2), 844–853. https://doi.org/10.1109/TGRS.2016.2616355 (2017).
54. Ding, Y. et al. Self-supervised locality preserving low-pass graph convolutional embedding for large-scale hyperspectral image clustering. *IEEE Trans. Geosci. Remote Sens.* **60**, 1–16. https://doi.org/10.1109/TGRS.2022.3198842 (2022).
55. He, Y. et al. Generation of 1 km high resolution Standardized Precipitation Evapotranspiration Index for drought monitoring over China using Google Earth Engine. *Int. J. Appl. Earth Observ. Geoinf.* **135**, 104296 (2024).
56. Yu, W., Zhou, P., Yan, S. et al. Inceptionnext: When inception meets convnext. in *Proceedings of the CVPR.5672-5683* (2024).
57. Liu, M., et al. CM-UNet: Hybrid CNN-Mamba UNet for remote sensing image semantic segmentation. Preprint at arXiv:2405.10530 (2024).
58. Sun, L. et al. Spatially-adaptive feature modulation for efficient image super-resolution. in *Proceedings of the ICCV* (2023).
59. Sun, L., Zhao, G., Zheng, Y. & Wu, Z. Spectral–spatial feature tokenization transformer for hyperspectral image classification. *IEEE Trans. Geosci. Remote Sens.* **60**, 1–14. https://doi.org/10.1109/TGRS.2022.3144158 (2022).
60. Roy, S. K. et al. Spectral–spatial morphological attention transformer for hyperspectral image classification. *IEEE Trans. Geosci. Remote Sens.* **61**, 1–15. https://doi.org/10.1109/TGRS.2023.3242346 (2023).
61. Zhao, Z., Xu, X., Li, S. & Plaza, A. Hyperspectral image classification using groupwise separable convolutional vision transformer network. *IEEE Trans. Geosci. Remote Sens.* **62**, 1–17. https://doi.org/10.1109/TGRS.2024.3377610 (2024).

Acknowledgements

This research was supported in part by the National Natural Science Foundation of China under Grant 62071175, the Hunan Provincial Natural Science Foundation of China under Grant 2024JJ8359, the Hunan Provincial Department of Education Scientific Research under Grant 22B0376, the Hunan Province Traditional Chinese Medicine Scientific Research Project under Grant A2024003, the 2022 Doctoral Research Initiation Fund of Hunan University of Chinese Medicine under Grant 0001036, the University-level Scientific Research Fund Project of Hunan University of Chinese Medicine under Grant 2024XJZA005.

Author contributions

Methodology, H.W. and J.L.; software,H.W, S.Q, Z.W, H.X. and Y.P.; investigation, J.L, H.L, Z.W, H.X. and Y.P.; validation ,H.W, S.Q, H.L, Z.W, H.X. and Y.P.;writing—original draft preparation, H. W.; writing—review and editing,S.Q, H.X., H.L, Z.W, and J.L. All authors have read and agreed to the published version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to J.L.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2025